Solving Protein-Ligand Accessibility Problems Using HAS-RRT

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BACKGROUND

- Motion planning is used to find a collision-free path through a space from a start configuration to a goal configuration.
- Navigates a configuration space that may contain obstacles.
- We can use RRT (Rapidly-Exploring Random Trees) to help solve motion planning problems.
- It is difficult to find a path when the obstacles form a narrow passage

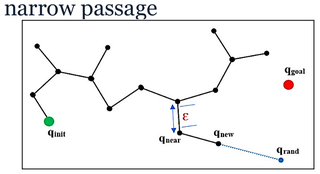


Figure 1. Simple example of RRT without obstacles. RRT grows a network of nodes out randomly to get from a starting point to a goal [4].

- Idea for HAS-RRT: We should be able to use topological guidance to aid sampling-based planning methods.
- A workspace skeleton is a graph that denotes the connectivity of a workspace.
- Use the workspace skeleton to provide guidance to the RRT [1].

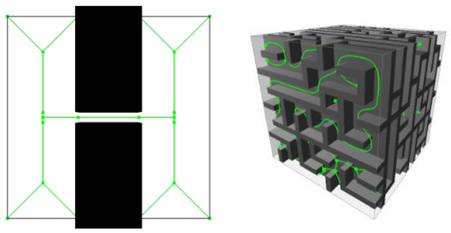


Figure 2. (left) Simple Passage environment, with a Medial Axis workspace skeleton.
(right) gridmaze8 environment, with a curated Mean Curvature workspace skeleton.

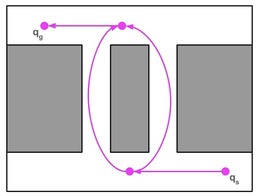
SUMMARY OF PREVIOUS WORK

- Hierarchical Annotated Skeleton RRT (HAS-RRT) can rapidly discover paths indicated by a workspace skeleton.
- Has the following features compared to other workspace-guided methods:
  - Fewer collision detection calls
  - Sparser roadmaps (fewer vertices and edges)
  - Faster than other methods

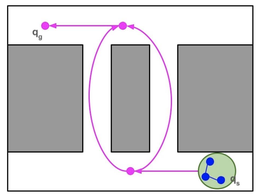
PREVIOUS WORK - METHOD

We present an RRT-based strategy that:

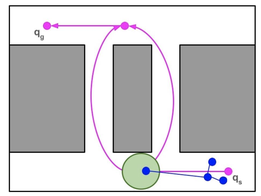
- Follows the edges of the workspace skeleton where possible.
- Takes smaller steps along the workspace skeleton when necessary.



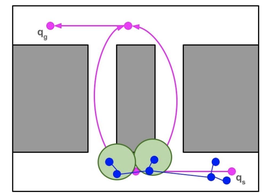
Step 1: Query Skeleton (purple) computed from a query q_s, q_g.



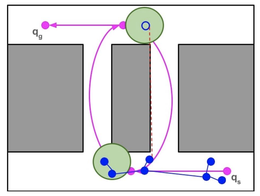
Step 2: Initial region (green) placed at the source vertex of the Query Skeleton.



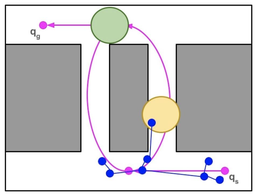
Step 3: A full extension to the end of the skeleton edge.



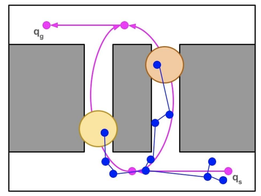
Step 4: Region splits into two regions at the adjacent edges



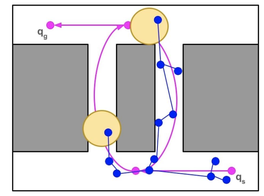
Step 5: An invalid extension is attempted.



Step 6: After a failed extension, a local expansion is attempted.



Step 7: The weight of the failed region is adjusted to bias the roadmap to alternative path options.



Step 8: The roadmap after a few iterations.

PREVIOUS WORK - RESULTS

- HAS-RRT was tested against 3 other methods [1, 2, 3], in 5 different environments. (Figures 2, 3)
- Runtime improvements:
  - 80+% speedup in Extended Sequence, Grid Tunnels, and Grid Mining environments.
  - 91% speedup in Grid Maze environment.
- Graph size improvements:
  - 75-90% improvement in #of vertices to DR-RRT. [2]
  - With a good skeleton, graph is visually sparser. (Figure 3)

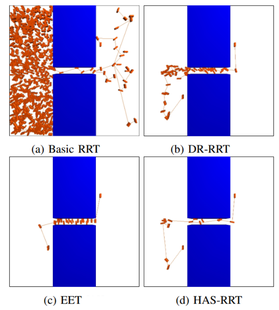


Figure 3. Example roadmaps from Simple Passage query.

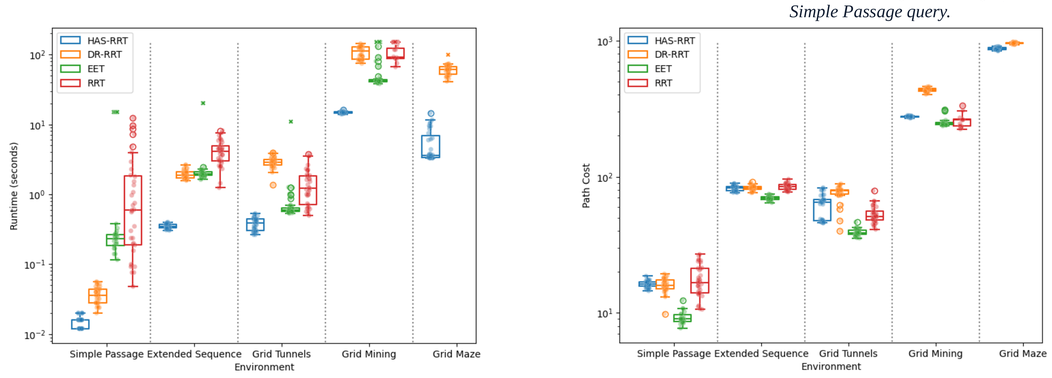


Figure 4. Runtimes (right) and Path Costs (length; left) for HAS-RRT and other comparison methods in 5 different environments.

OUR PROJECT

- Protein-ligand interactions are essential to many biochemical processes where a molecule needs to bind to a protein.
- Goal: find a valid path for a ligand to a binding site on a protein.
- Motion planning algorithms from robotics can help find this path by treating the ligand as a robot to find a valid path to the protein, which is treated as an obstacle.
- We are using HAS-RRT to solve this problem and find a low energy path to the binding site, since energy needs to be conserved in biochemical reactions.

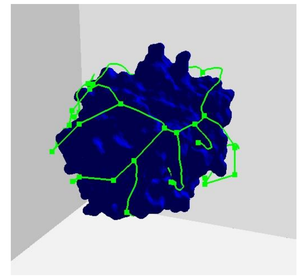


Figure 5: DhaA protein (blue), with a mean curvature workspace skeleton (green).

REFERENCES

[1] RRT: S. M. LaValle and J. J. Kuffner, "Rapidly-exploring random trees: Progress and prospects," in *New Directions in Algorithmic and Computational Robotics*. A. K. Peters, 2001, pp. 293–308.
[2] DR-RRT: R. Sandström, D. Uwacu, J. Denny and N. M. Amato, "Topology-Guided Roadmap Construction With Dynamic Region Sampling," in *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 6161-6168, Oct. 2020, doi: 10.1109/LRA.2020.3010487.
[3] EET: M. Rickert, O. Brock and A. Knoll, "Balancing exploration and exploitation in motion planning," *2008 IEEE International Conference on Robotics and Automation*, Pasadena, CA, USA, 2008, pp. 2812-2817, doi: 10.1109/ROBOT.2008.4543636.
[4] S. Al-Ansarry and S. Al-Darraji. "Hybrid RRT-A*: An Improved Path Planning Method for an Autonomous Mobile Robots." *Iraqi Journal for Electrical & Electronic Engineering 17.1 (2021).*

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